CYBERSRS

Secure Backup and Recovery Management System

Software Requirements Specification

DRAFT — human review required

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AI-generated advisory document. Review by a qualified professional is required.

Automated checks do not prove completeness or semantic correctness. SDLC suitability scores are rule-based points, not probabilities.

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1. Introduction

Original project description

Design a secure backup and recovery management system for a university that protects critical data from academic departments, administrative systems, research platforms, and shared services. The system should allow authorized administrators to define backup policies, initiate and monitor backup jobs, restore approved data, and review backup status across multiple campus locations. It should detect failed or incomplete backup operations, generate security and operational alerts, maintain audit logs of backup and restore activities, and restrict backup administration based on user roles. The platform should integrate with existing university storage and identity services and remain secure, reliable, scalable, and manageable.

Summary: This specification records the project purpose, stakeholder constraints and unresolved decisions stated in the source evidence below.

Items

Stakeholder source statement E-001

Detail: Design a secure backup and recovery management system for a university that protects critical data from academic departments, administrative systems, research platforms, and shared services.

Status: confirmed

Evidence ids: E-001

Stakeholder source statement E-002

Detail: The system should allow authorized administrators to define backup policies, initiate and monitor backup jobs, restore approved data, and review backup status across multiple campus locations.

Status: confirmed

Evidence ids: E-002

Stakeholder source statement E-003

Detail: It should detect failed or incomplete backup operations, generate security and operational alerts, maintain audit logs of backup and restore activities, and restrict backup administration based on user roles.

Status: confirmed

Evidence ids: E-003

Stakeholder source statement E-004

Detail: The platform should integrate with existing university storage and identity services and remain secure, reliable, scalable, and manageable.

Status: confirmed

Evidence ids: E-004

2. Functional requirements

Applicability reason: Independent source obligations; unspecified choices remain TBD.

Requirements

FR-001 - protect critical data from academic departments, administrative systems, research platforms, and shared services

Statement: The system shall protect critical data from academic departments, administrative systems, research platforms, and shared services.

Rationale: Source obligation E-001_O01.

Priority: must

Dimensions: business

Basis: explicit

Evidence ids: E-001

Acceptance criteria

Given: the system protects critical data from academic departments, administrative systems, research platforms, and shared services

When: the system initiates a protection operation for designated data assets

Then: the protected data remains accessible and intact without unauthorized modification or loss

Method: test

Observable evidence: External test capture of the resulting state

FR-002 - allow authorized administrators to define backup policies

Statement: The system shall allow authorized administrators to define backup policies.

Rationale: Source obligation E-002_O01.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: an administrator defines a backup policy for a data set

When: the administrator initiates the policy definition operation

Then: the named policy is available for examination in the system interface

Method: test

Observable evidence: External test capture of the resulting response

FR-003 - allow authorized administrators to initiate backup jobs

Statement: The system shall allow authorized administrators to initiate backup jobs.

Rationale: Source obligation E-002_O02.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: an administrator initiates a backup job for a data set

When: the administrator starts the backup job operation

Then: the named job enters its active state and is visible in the system interface

Method: test

Observable evidence: External test capture of the resulting response

FR-004 - allow authorized administrators to monitor backup jobs

Statement: The system shall allow authorized administrators to monitor backup jobs.

Rationale: Source obligation E-002_O03.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: an administrator monitors a running backup job

When: the administrator performs a monitoring action on the job

Then: the system returns status updates for the named job

Method: inspection

Observable evidence: External test capture of the resulting state

FR-005 - allow authorized administrators to restore approved data

Statement: The system shall allow authorized administrators to restore approved data.

Rationale: Source obligation E-002_O04.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: an administrator restores a data set from a backup copy

When: the administrator executes the restore operation

Then: the named data or configuration is restored to its expected state and is accessible in the system interface

Method: test

Observable evidence: External test capture of the resulting response

FR-006 - allow authorized administrators to review backup status across multiple campus locations

Statement: The system shall allow authorized administrators to review backup status across multiple campus locations.

Rationale: Source obligation E-002_O05.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: an administrator requests backup status for multiple campus locations

When: the administrator performs a cross-location status query

Then: the named items are returned in the requested format and are available for examination

Method: demonstration

Observable evidence: External test capture of the inspection record

FR-007 - detect failed or incomplete backup operations

Statement: The system shall detect failed or incomplete backup operations.

Rationale: Source obligation E-003_O01.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-003

Acceptance criteria

Given: system detects a failed or incomplete backup operation

When: the system initiates detection of a failed or incomplete backup operation

Then: the system reports the failure with identifiable metadata

Method: test

Observable evidence: External test capture of the resulting response

FR-008 - generate security and operational alerts

Statement: The system shall generate security and operational alerts.

Rationale: Source obligation E-003_O02.

Priority: should

Dimensions: business; security

Basis: explicit

Evidence ids: E-003

Acceptance criteria

Given: system identifies an alert condition for security or operational risk

When: the system initiates the alert condition

Then: the requested alert exists in the designated channel

Method: inspection

Observable evidence: External test capture of the resulting state

FR-009 - maintain audit logs of backup and restore activities

Statement: The system shall maintain audit logs of backup and restore activities.

Rationale: Source obligation E-003_O03.

Priority: should

Dimensions: business; auditability; data

Basis: explicit

Evidence ids: E-003

Acceptance criteria

Given: system records backup and restore activities

When: a backup or restore operation is completed

Then: The system records backup and restore activities when a backup or restore operation is completed, as evidenced by external test capture of the inspection record.

Method: analysis

Observable evidence: External test capture of the inspection record

Unresolved decisions: Decision Q-3aae42e94793 is TBD.; Decision Q-64ef2e28335c is TBD.

FR-010 - restrict backup administration based on user roles

Statement: The system shall restrict backup administration based on user roles.

Rationale: Source obligation E-003_O04.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-003

Acceptance criteria

Given: user attempts to perform backup administration

When: the user initiates an administrative action

Then: the system denies access if the user role lacks backup administration privileges

Method: demonstration

Observable evidence: External test capture of the resulting state

FR-011 - integrate with existing university storage and identity services

Statement: The system shall integrate with existing university storage and identity services.

Rationale: Source obligation E-004_O01.

Priority: should

Dimensions: business; integration

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: system integrates with existing university storage and identity services using a controlled exchange

When: the system initiates a connection request to the external service

Then: the external service responds with an authenticated session token and confirms successful path establishment

Method: test

Observable evidence: External test capture of the resulting response

3. Non-functional requirements

Applicability reason: Independent source obligations; unspecified choices remain TBD.

Requirements

NFR-001 - be secure

Statement: The system shall be secure.

Rationale: Source obligation E-001_O02.

Priority: must

Dimensions: security

Basis: explicit

Evidence ids: E-001

Acceptance criteria

Given: A draft acceptance plan for this quality goal: the system shall be secure against unauthorized access and data leakage during backup and recovery operations.

When: An analyst inspects or reviews that acceptance plan to determine whether measurable targets are defined or remain TBD/uncertain.

Then: The plan's measurable targets remain TBD or unconfirmed, as no specific metrics or thresholds for security assurance are provided in the current draft.

Method: inspection

Observable evidence: External test capture of the inspection record

Unresolved decisions: Decision Q-7b2e5f89a0b1 is TBD.

NFR-002 - remain secure

Statement: The system shall remain secure.

Rationale: Source obligation E-004_O02.

Priority: should

Dimensions: security

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: A draft acceptance plan for this quality goal: the system shall remain secure during backup and recovery operations.

When: An analyst inspects or reviews that acceptance plan to determine whether it includes measurable targets for security assurance.

Then: The plan's measurable targets remain TBD or unconfirmed; no specific metric, threshold, or test case is defined in the current draft.

Method: inspection

Observable evidence: External test capture of the resulting response

Unresolved decisions: Decision Q-7b2e5f89a0b1 is TBD.

NFR-003 - remain reliable

Statement: The system shall remain reliable.

Rationale: Source obligation E-004_O03.

Priority: should

Dimensions: availability

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: A draft acceptance plan for this quality goal: the system shall remain reliable during backup and recovery operations.

When: An analyst inspects or reviews that acceptance plan to determine whether it includes measurable targets for reliability assurance.

Then: The plan's measurable targets remain TBD or unconfirmed; no specific metric, threshold, or test case is defined in the current draft.

Method: inspection

Observable evidence: External test capture of the resulting response

Unresolved decisions: Decision Q-819bb986f419 is TBD.

NFR-004 - remain scalable

Statement: The system shall remain scalable.

Rationale: Source obligation E-004_O04.

Priority: should

Dimensions: performance

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: A draft acceptance plan for this quality goal: the system shall remain scalable during backup and recovery operations.

When: An analyst inspects or reviews that acceptance plan to determine whether it includes measurable targets for scalability assurance.

Then: The plan's measurable targets remain TBD or unconfirmed; no specific metric, threshold, or test case is defined in the current draft.

Method: inspection

Observable evidence: External test capture of the resulting state

Unresolved decisions: Decision Q-543394346815 is TBD.

NFR-005 - remain manageable

Statement: The system shall remain manageable.

Rationale: Source obligation E-004_O05.

Priority: should

Dimensions: usability

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: A draft acceptance plan for this quality goal: the system shall remain manageable during backup and recovery operations.

When: An analyst inspects or reviews that acceptance plan to determine whether it includes measurable targets for manageability assurance.

Then: The plan's measurable targets remain TBD or unconfirmed; no specific metric, threshold, or test case is defined in the current draft.

Method: inspection

Observable evidence: External test capture of the inspection record

Unresolved decisions: Decision Q-5268b96e41b0 is TBD.

Related requirement ids: FR-001; FR-011

4. Security, privacy and compliance

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: FR-002; FR-003; FR-004; FR-005; FR-006; FR-007; FR-008; FR-009; FR-010; NFR-001; NFR-002

5. Data requirements

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: FR-007; FR-008; FR-009; FR-010; FR-011

6. Network and interfaces

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: FR-011; NFR-002; NFR-003; NFR-004; NFR-005

7. Compliance and policy applicability matrix

Open decisions: Decision Q-66cf5513b374 is TBD.; Decision Q-dad9a79bf709 is TBD.

8. Architecture

Summary: Proposed logical components and interactions mapped to canonical source obligations.

Items

Actor: authorized administrators

Detail: Proposed logical endpoint for the source-stated actor: authorized administrators.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-002; FR-003; FR-004; FR-005; FR-006

Component name: Actor: authorized administrators

Responsibility: Represent the source-stated actor: authorized administrators.

Backup Administration Interface

Detail: Proposed logical owner for FR-002, FR-003, FR-004, FR-005, FR-006, FR-001, FR-010, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-002; E-003; E-004

Requirement ids: FR-002; FR-003; FR-004; FR-005; FR-006; FR-001; FR-010; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

Component name: Backup Administration Interface

Responsibility: The system shall allow authorized administrators to define backup policies. The system shall allow authorized administrators to initiate backup jobs. The system shall allow authorized administrators to monitor backup jobs. The system shall allow authorized administrators to restore approved data. The system shall allow authorized administrators to review backup status across multiple campus locations. The system shall protect critical data from academic departments, administrative systems, research platforms, and shared services. The system shall restrict backup administration based on user roles. The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Security Monitoring Engine

Detail: Proposed logical owner for FR-007, FR-008, FR-001, FR-010, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-003; E-004

Requirement ids: FR-007; FR-008; FR-001; FR-010; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

Component name: Security Monitoring Engine

Responsibility: The system shall detect failed or incomplete backup operations. The system shall generate security and operational alerts. The system shall protect critical data from academic departments, administrative systems, research platforms, and shared services. The system shall restrict backup administration based on user roles. The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Audit and Logging Service

Detail: Proposed logical owner for FR-009, FR-001, FR-010, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-003; E-004

Requirement ids: FR-009; FR-001; FR-010; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

Component name: Audit and Logging Service

Responsibility: The system shall maintain audit logs of backup and restore activities. The system shall protect critical data from academic departments, administrative systems, research platforms, and shared services. The system shall restrict backup administration based on user roles. The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Integration Layer

Detail: Proposed logical owner for FR-011.

Status: proposed

Evidence ids: E-004

Requirement ids: FR-011

Component name: Integration Layer

Responsibility: The system shall integrate with existing university storage and identity services.

Data flows

Proposed source-defined interaction

Detail: Proposed directed path for FR-002, FR-003, FR-004, FR-005, FR-006.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-002; FR-003; FR-004; FR-005; FR-006

From component: Actor: authorized administrators

To component: Backup Administration Interface

Data or action: The system shall allow authorized administrators to define backup policies. The system shall allow authorized administrators to initiate backup jobs. The system shall allow authorized administrators to monitor backup jobs. The system shall allow authorized administrators to restore approved data. The system shall allow authorized administrators to review backup status across multiple campus locations.

Proposed source-defined interaction

Detail: Proposed directed path for FR-001, FR-010, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-003; E-004

Requirement ids: FR-001; FR-010; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

From component: Backup Administration Interface

To component: Security Monitoring Engine

Data or action: The system shall protect critical data from academic departments, administrative systems, research platforms, and shared services. The system shall restrict backup administration based on user roles. The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Proposed source-defined interaction

Detail: Proposed directed path for FR-001, FR-010, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-003; E-004

Requirement ids: FR-001; FR-010; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

From component: Backup Administration Interface

To component: Audit and Logging Service

Data or action: The system shall protect critical data from academic departments, administrative systems, research platforms, and shared services. The system shall restrict backup administration based on user roles. The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Proposed source-defined interaction

Detail: Proposed directed path for FR-001, FR-010, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-003; E-004

Requirement ids: FR-001; FR-010; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

From component: Security Monitoring Engine

To component: Backup Administration Interface

Data or action: The system shall protect critical data from academic departments, administrative systems, research platforms, and shared services. The system shall restrict backup administration based on user roles. The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Proposed source-defined interaction

Detail: Proposed directed path for FR-001, FR-010, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-003; E-004

Requirement ids: FR-001; FR-010; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

From component: Security Monitoring Engine

To component: Audit and Logging Service

Data or action: The system shall protect critical data from academic departments, administrative systems, research platforms, and shared services. The system shall restrict backup administration based on user roles. The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Proposed source-defined interaction

Detail: Proposed directed path for FR-001, FR-010, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-003; E-004

Requirement ids: FR-001; FR-010; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

From component: Audit and Logging Service

To component: Backup Administration Interface

Data or action: The system shall protect critical data from academic departments, administrative systems, research platforms, and shared services. The system shall restrict backup administration based on user roles. The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Proposed source-defined interaction

Detail: Proposed directed path for FR-001, FR-010, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-003; E-004

Requirement ids: FR-001; FR-010; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

From component: Audit and Logging Service

To component: Security Monitoring Engine

Data or action: The system shall protect critical data from academic departments, administrative systems, research platforms, and shared services. The system shall restrict backup administration based on user roles. The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Trust boundaries

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for FR-002, FR-003, FR-004, FR-005, FR-006.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-002; FR-003; FR-004; FR-005; FR-006

Untrusted component: Actor: authorized administrators

Trusted component: Backup Administration Interface

Enforcement: The system shall allow authorized administrators to define backup policies. The system shall allow authorized administrators to initiate backup jobs. The system shall allow authorized administrators to monitor backup jobs. The system shall allow authorized administrators to restore approved data. The system shall allow authorized administrators to review backup status across multiple campus locations.

9. Threats

Threats

Id: THR-001

Asset: User identities and authenticated access

Actor: Hypothetical attacker, privileged insider or failure cause

Scenario: Derived hypothesis: An attacker uses a compromised identity or authenticated session to impersonate an authorized user and exercise that user's permitted operations.

Impact: Unauthorized use of a legitimate identity could expose resources or alter system state.

Control requirement ids: FR-011; NFR-002; NFR-003; NFR-004; NFR-005

Evidence ids: E-004

Residual risk: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

Id: THR-002

Asset: Role and permission boundaries

Actor: Hypothetical attacker, privileged insider or failure cause

Scenario: Derived hypothesis: A principal exploits an incorrect role assignment or permission check to perform operations outside the source-defined authorization boundary.

Impact: A lower-privilege user could access restricted functions or sensitive resources.

Control requirement ids: FR-002; FR-003; FR-004; FR-005; FR-006; FR-007; FR-008; FR-009; FR-010

Evidence ids: E-002; E-003

Residual risk: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

Id: THR-003

Asset: Administrative security decisions

Actor: Hypothetical attacker, privileged insider or failure cause

Scenario: Derived hypothesis: A malicious or compromised administrator misuses a source-permitted management operation to weaken protection or disrupt legitimate use through the stated administrative workflow.

Impact: Misused administrative authority could undermine protection or interrupt legitimate use.

Control requirement ids: FR-002; FR-003; FR-004; FR-005; FR-006

Evidence ids: E-002

Residual risk: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

Id: THR-004

Asset: Integration trust paths

Actor: Hypothetical attacker, privileged insider or failure cause

Scenario: Derived hypothesis: A compromised connected system or integration path supplies forged, altered or missing identity, event or policy data used by the source-defined integration.

Impact: Untrusted integration data could cause incorrect decisions or hide security events.

Control requirement ids: FR-011; NFR-002; NFR-003; NFR-004; NFR-005

Evidence ids: E-004

Residual risk: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

Id: THR-005

Asset: Monitoring and security alerts

Actor: Hypothetical attacker, privileged insider or failure cause

Scenario: Derived hypothesis: Suspicious activity evades the source-defined detection workflow, or excessive events obscure actionable alerts before an authorized reviewer can investigate.

Impact: Relevant suspicious activity could remain undetected or receive delayed investigation.

Control requirement ids: FR-002; FR-003; FR-004; FR-005; FR-006; FR-007; FR-008; FR-009; FR-010

Evidence ids: E-002; E-003

Residual risk: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

Id: THR-006

Asset: Security and administrative audit evidence

Actor: Hypothetical attacker, privileged insider or failure cause

Scenario: Derived hypothesis: An attacker or recording failure suppresses or alters source-required audit evidence so that access decisions, security events or admin actions cannot be reconstructed.

Impact: Investigators could lose reliable evidence of the events required by the brief.

Control requirement ids: FR-007; FR-008; FR-009; FR-010

Evidence ids: E-003

Residual risk: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

Assumptions

1. These are defensive planning hypotheses, not observed incidents or approved extra controls. Unrecognized threats and actual implementation effectiveness require human review.

10. Risks

Risks

Id: RISK-001

Description: Derived risk affecting User identities and authenticated access: Unauthorized use of a legitimate identity could expose resources or alter system state.

Likelihood: unknown

Impact: unknown

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: An attacker uses a compromised identity or authenticated session to impersonate an authorized user and exercise that user's permitted operations.

Mitigation: Review impersonation and compromised-identity paths against the stated access duties; confirm any additional identity or session protections with the stakeholder. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Requirement ids: FR-011; NFR-002; NFR-003; NFR-004; NFR-005

Evidence ids: E-004

Owner: TBD

Id: RISK-002

Description: Derived risk affecting Role and permission boundaries: A lower-privilege user could access restricted functions or sensitive resources.

Likelihood: unknown

Impact: unknown

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: A principal exploits an incorrect role assignment or permission check to perform operations outside the source-defined authorization boundary.

Mitigation: Test permitted and denied role/resource pairs against the linked authorization duties; review ambiguous permission assignments with the resource owner. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Requirement ids: FR-002; FR-003; FR-004; FR-005; FR-006; FR-007; FR-008; FR-009; FR-010

Evidence ids: E-002; E-003

Owner: TBD

Id: RISK-003

Description: Derived risk affecting Administrative security decisions: Misused administrative authority could undermine protection or interrupt legitimate use.

Likelihood: unknown

Impact: unknown

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: A malicious or compromised administrator misuses a source-permitted management operation to weaken protection or disrupt legitimate use through the stated administrative workflow.

Mitigation: Review misuse cases for each source-permitted administrative operation and compare its outcomes and audit coverage with the stated management boundary. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Requirement ids: FR-002; FR-003; FR-004; FR-005; FR-006

Evidence ids: E-002

Owner: TBD

Id: RISK-004

Description: Derived risk affecting Integration trust paths: Untrusted integration data could cause incorrect decisions or hide security events.

Likelihood: unknown

Impact: unknown

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: A compromised connected system or integration path supplies forged, altered or missing identity, event or policy data used by the source-defined integration.

Mitigation: Review trust assumptions at the specified integration boundaries with missing or altered input fixtures; confirm the required failure handling with the owner. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Requirement ids: FR-011; NFR-002; NFR-003; NFR-004; NFR-005

Evidence ids: E-004

Owner: TBD

Id: RISK-005

Description: Derived risk affecting Monitoring and security alerts: Relevant suspicious activity could remain undetected or receive delayed investigation.

Likelihood: unknown

Impact: unknown

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: Suspicious activity evades the source-defined detection workflow, or excessive events obscure actionable alerts before an authorized reviewer can investigate.

Mitigation: Evaluate the stated detection duties with representative positive and negative events; confirm undecided thresholds, alert priorities and investigation responsibilities. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Requirement ids: FR-002; FR-003; FR-004; FR-005; FR-006; FR-007; FR-008; FR-009; FR-010

Evidence ids: E-002; E-003

Owner: TBD

Id: RISK-006

Description: Derived risk affecting Security and administrative audit evidence: Investigators could lose reliable evidence of the events required by the brief.

Likelihood: unknown

Impact: unknown

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: An attacker or recording failure suppresses or alters source-required audit evidence so that access decisions, security events or admin actions cannot be reconstructed.

Mitigation: Compare source-required events with the resulting audit evidence; assess missing and altered-record cases and confirm unprovided protection or retention decisions. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Requirement ids: FR-007; FR-008; FR-009; FR-010

Evidence ids: E-003

Owner: TBD

11. User stories and use cases

User stories and use cases

Id: UC-001

Actor: authorized administrators

Goal: The system shall allow authorized administrators to define backup policies.

Story: As authorized administrators, I need the following behavior: The system shall allow authorized administrators to define backup policies.

Preconditions

1. an administrator defines a backup policy for a data set

Steps

1. the administrator initiates the policy definition operation

Success outcome: the named policy is available for examination in the system interface

Requirement ids: FR-002

Evidence ids: E-002

Id: UC-002

Actor: authorized administrators

Goal: The system shall allow authorized administrators to initiate backup jobs.

Story: As authorized administrators, I need the following behavior: The system shall allow authorized administrators to initiate backup jobs.

Preconditions

1. an administrator initiates a backup job for a data set

Steps

1. the administrator starts the backup job operation

Success outcome: the named job enters its active state and is visible in the system interface

Requirement ids: FR-003

Evidence ids: E-002

Id: UC-003

Actor: authorized administrators

Goal: The system shall allow authorized administrators to monitor backup jobs.

Story: As authorized administrators, I need the following behavior: The system shall allow authorized administrators to monitor backup jobs.

Preconditions

1. an administrator monitors a running backup job

Steps

1. the administrator performs a monitoring action on the job

Success outcome: the system returns status updates for the named job

Requirement ids: FR-004

Evidence ids: E-002

Id: UC-004

Actor: authorized administrators

Goal: The system shall allow authorized administrators to restore approved data.

Story: As authorized administrators, I need the following behavior: The system shall allow authorized administrators to restore approved data.

Preconditions

1. an administrator restores a data set from a backup copy

Steps

1. the administrator executes the restore operation

Success outcome: the named data or configuration is restored to its expected state and is accessible in the system interface

Requirement ids: FR-005

Evidence ids: E-002

Id: UC-005

Actor: authorized administrators

Goal: The system shall allow authorized administrators to review backup status across multiple campus locations.

Story: As authorized administrators, I need the following behavior: The system shall allow authorized administrators to review backup status across multiple campus locations.

Preconditions

1. an administrator requests backup status for multiple campus locations

Steps

1. the administrator performs a cross-location status query

Success outcome: the named items are returned in the requested format and are available for examination

Requirement ids: FR-006

Evidence ids: E-002

12. Verification and acceptance

Tests

Id: TEST-001

Requirement ids: FR-001

Procedure

1. Given: the system protects critical data from academic departments, administrative systems, research platforms, and shared services
2. When: the system initiates a protection operation for designated data assets

Expected result: Expected: the protected data remains accessible and intact without unauthorized modification or loss

Evidence to record: Evidence: External test capture of the resulting state

Test type: positive

Id: TEST-002

Requirement ids: FR-002

Procedure

1. Given: an administrator defines a backup policy for a data set
2. When: the administrator initiates the policy definition operation

Expected result: Expected: the named policy is available for examination in the system interface

Evidence to record: Evidence: External test capture of the resulting response

Test type: positive

Id: TEST-003

Requirement ids: FR-003

Procedure

1. Given: an administrator initiates a backup job for a data set
2. When: the administrator starts the backup job operation

Expected result: Expected: the named job enters its active state and is visible in the system interface

Evidence to record: Evidence: External test capture of the resulting response

Test type: positive

Id: TEST-004

Requirement ids: FR-004

Procedure

1. Given: an administrator monitors a running backup job
2. When: the administrator performs a monitoring action on the job

Expected result: Expected: the system returns status updates for the named job

Evidence to record: Evidence: External test capture of the resulting state

Test type: inspection

Id: TEST-005

Requirement ids: FR-005

Procedure

1. Given: an administrator restores a data set from a backup copy
2. When: the administrator executes the restore operation

Expected result: Expected: the named data or configuration is restored to its expected state and is accessible in the system interface

Evidence to record: Evidence: External test capture of the resulting response

Test type: positive

Id: TEST-006

Requirement ids: FR-006

Procedure

1. Given: an administrator requests backup status for multiple campus locations
2. When: the administrator performs a cross-location status query

Expected result: Expected: the named items are returned in the requested format and are available for examination

Evidence to record: Evidence: External test capture of the inspection record

Test type: positive

Id: TEST-007

Requirement ids: FR-007

Procedure

1. Given: system detects a failed or incomplete backup operation
2. When: the system initiates detection of a failed or incomplete backup operation

Expected result: Expected: the system reports the failure with identifiable metadata

Evidence to record: Evidence: External test capture of the resulting response

Test type: positive

Id: TEST-008

Requirement ids: FR-008

Procedure

1. Given: system identifies an alert condition for security or operational risk
2. When: the system initiates the alert condition

Expected result: Expected: the requested alert exists in the designated channel

Evidence to record: Evidence: External test capture of the resulting state

Test type: inspection

Id: TEST-009

Requirement ids: FR-009

Procedure

1. Given: system records backup and restore activities
2. When: a backup or restore operation is completed

Expected result: Expected: The system records backup and restore activities when a backup or restore operation is completed, as evidenced by external test capture of the inspection record.

Evidence to record: Evidence: External test capture of the inspection record

Test type: analysis

Id: TEST-010

Requirement ids: FR-010

Procedure

1. Given: user attempts to perform backup administration
2. When: the user initiates an administrative action

Expected result: Expected: the system denies access if the user role lacks backup administration privileges

Evidence to record: Evidence: External test capture of the resulting state

Test type: negative

Id: TEST-011

Requirement ids: FR-011

Procedure

1. Given: system integrates with existing university storage and identity services using a controlled exchange
2. When: the system initiates a connection request to the external service

Expected result: Expected: the external service responds with an authenticated session token and confirms successful path establishment

Evidence to record: Evidence: External test capture of the resulting response

Test type: positive

Id: TEST-012

Requirement ids: NFR-001

Procedure

1. Given: A draft acceptance plan for this quality goal: the system shall be secure against unauthorized access and data leakage during backup and recovery operations.
2. When: An analyst inspects or reviews that acceptance plan to determine whether measurable targets are defined or remain TBD/uncertain.

Expected result: Expected: The plan's measurable targets remain TBD or unconfirmed, as no specific metrics or thresholds for security assurance are provided in the current draft.

Evidence to record: Evidence: External test capture of the inspection record

Test type: inspection

Id: TEST-013

Requirement ids: NFR-002

Procedure

1. Given: A draft acceptance plan for this quality goal: the system shall remain secure during backup and recovery operations.
2. When: An analyst inspects or reviews that acceptance plan to determine whether it includes measurable targets for security assurance.

Expected result: Expected: The plan's measurable targets remain TBD or unconfirmed; no specific metric, threshold, or test case is defined in the current draft.

Evidence to record: Evidence: External test capture of the resulting response

Test type: inspection

Id: TEST-014

Requirement ids: NFR-003

Procedure

1. Given: A draft acceptance plan for this quality goal: the system shall remain reliable during backup and recovery operations.
2. When: An analyst inspects or reviews that acceptance plan to determine whether it includes measurable targets for reliability assurance.

Expected result: Expected: The plan's measurable targets remain TBD or unconfirmed; no specific metric, threshold, or test case is defined in the current draft.

Evidence to record: Evidence: External test capture of the resulting response

Test type: inspection

Id: TEST-015

Requirement ids: NFR-004

Procedure

1. Given: A draft acceptance plan for this quality goal: the system shall remain scalable during backup and recovery operations.
2. When: An analyst inspects or reviews that acceptance plan to determine whether it includes measurable targets for scalability assurance.

Expected result: Expected: The plan's measurable targets remain TBD or unconfirmed; no specific metric, threshold, or test case is defined in the current draft.

Evidence to record: Evidence: External test capture of the resulting state

Test type: inspection

Id: TEST-016

Requirement ids: NFR-005

Procedure

1. Given: A draft acceptance plan for this quality goal: the system shall remain manageable during backup and recovery operations.
2. When: An analyst inspects or reviews that acceptance plan to determine whether it includes measurable targets for manageability assurance.

Expected result: Expected: The plan's measurable targets remain TBD or unconfirmed; no specific metric, threshold, or test case is defined in the current draft.

Evidence to record: Evidence: External test capture of the inspection record

Test type: inspection

Limitations: These verification plans have not been executed; their acceptance outcomes require testing.

13. Glossary

Summary: Source-present terminology; definitions introduce no additional product obligation.

Items

Access restoration

Detail: Returning access that was previously restricted, within the source-defined permission boundary.

Status: proposed

Evidence ids: E-002; E-003

Log

Detail: A collection of event records. This term does not itself require additional logging fields, events or access permissions.

Status: proposed

Evidence ids: E-003

Audit record

Detail: A stored record of an event. Required event scope and retention are determined by the linked source, not this definition.

Status: proposed

Evidence ids: E-003

Policy

Detail: A rule or decision governing permitted actions; its project-specific meaning remains defined by the source requirements.

Status: proposed

Evidence ids: E-002

14. SDLC recommendation and workflow

Selected model: Incremental

Justification: The Incremental model is selected based on derived evidence showing the need for integrated security review (E-002), operational validation through monitoring feedback (E-002, E-003), staged deployment across multiple locations (E-002), and integration verification of connected systems (E-004). These factors collectively support a phased, iterative approach that enables continuous assurance without assuming continuous deployment or full-scale rollout. The Incremental model scores highest at 66 points—outperforming Hybrid (65) and DevSecOps (63)—due to superior alignment with operational validation and deployment scope requirements. While security assurance is well-supported by all models, the derived operational feedback and staged rollout opportunities are most effectively addressed in an incremental lifecycle. The Hybrid model lacks sufficient weighting on deployment scope and integration assurance, while DevSecOps underweights deployment scope and overemphasizes security at the expense of operational validation. Spiral, Agile, and V-Model scores are significantly lower due to absence of differentiating evidence. Unknown factors such as budget, failure consequences, legacy systems, and verification methods remain unaddressed, making this recommendation provisional. The decision status explicitly notes that delivery constraints and team capacity must be confirmed before adoption.

Tradeoffs

1. Recurring delivery requires stakeholder review and backlog refinement; a cycle does not guarantee completeness or remove unresolved decisions.
2. Proposed suitability recommendation; confirm delivery constraints and team capacity before adopting this lifecycle. Missing factors are not assumed.

Phases

Name: Requirements and verification planning

Phase type: planning

Activities

1. Review and baseline the source-defined work group: FR-001: The system shall protect critical data from academic departments, administrative systems, research platforms, and shared services.; FR-002: The system shall allow authorized administrators to define backup policies.; FR-003: The system shall allow authorized administrators to initiate backup jobs.; FR-004: The system shall allow authorized administrators to monitor backup jobs.; FR-005: The system shall allow authorized administrators to restore approved data.; FR-006: The system shall allow authorized administrators to review backup status across multiple campus locations.; FR-007: The system shall detect failed or incomplete backup operations.; FR-008: The system shall generate security and operational alerts.; FR-009: The system shall maintain audit logs of backup and restore activities.; FR-010: The system shall restrict backup administration based on user roles.; FR-011: The system shall integrate with existing university storage and identity services.; NFR-001: The system shall be secure.; NFR-002: The system shall remain secure.; NFR-003: The system shall remain reliable.; NFR-004: The system shall remain scalable.; NFR-005: The system shall remain manageable.
2. Create the unit, integration, system and acceptance test plan before implementation; retain each authored acceptance criterion.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Versioned requirements baseline
2. Unit, integration, system and acceptance test plan
3. Unresolved-decision record

Entry criteria

1. Authoritative inputs and canonical requirements are available for review.

Exit criteria

1. A human reviewer has recorded the baseline decision and disposition of unresolved scope.

Security checks

1. Review implementation and acceptance evidence against FR-008: The system shall generate security and operational alerts.
2. Review implementation and acceptance evidence against NFR-001: The system shall be secure.
3. Review implementation and acceptance evidence against NFR-002: The system shall remain secure.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; FR-004; FR-005; FR-006; FR-007; FR-008; FR-009; FR-010; FR-011; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

Name: Incremental delivery cycle

Phase type: iteration

Activities

1. Review and prioritize the backlog for the next iteration without changing source-defined scope.
2. Implement the source-defined work group: FR-001: The system shall protect critical data from academic departments, administrative systems, research platforms, and shared services.; FR-002: The system shall allow authorized administrators to define backup policies.; FR-003: The system shall allow authorized administrators to initiate backup jobs.; FR-004: The system shall allow authorized administrators to monitor backup jobs.; FR-005: The system shall allow authorized administrators to restore approved data.; FR-006: The system shall allow authorized administrators to review backup status across multiple campus locations.; FR-007: The system shall detect failed or incomplete backup operations.; FR-008: The system shall generate security and operational alerts.; FR-009: The system shall maintain audit logs of backup and restore activities.; FR-010: The system shall restrict backup administration based on user roles.; FR-011: The system shall integrate with existing university storage and identity services.; NFR-001: The system shall be secure.; NFR-002: The system shall remain secure.; NFR-003: The system shall remain reliable.; NFR-004: The system shall remain scalable.; NFR-005: The system shall remain manageable.
3. In an authorized isolated test environment, execute TEST-001: Given: the system protects critical data from academic departments, administrative systems, research platforms, and shared services When: the system initiates a protection operation for designated data assets Expected: the protected data remains accessible and intact without unauthorized modification or loss
4. In an authorized isolated test environment, execute TEST-002: Given: an administrator defines a backup policy for a data set When: the administrator initiates the policy definition operation Expected: the named policy is available for examination in the system interface

5. In an authorized isolated test environment, execute TEST-003: Given: an administrator initiates a backup job for a data set When: the administrator starts the backup job operation Expected: the named job enters its active state and is visible in the system interface
6. In an authorized isolated test environment, execute TEST-004: Given: an administrator monitors a running backup job When: the administrator performs a monitoring action on the job Expected: the system returns status updates for the named job
7. In an authorized isolated test environment, execute TEST-005: Given: an administrator restores a data set from a backup copy When: the administrator executes the restore operation Expected: the named data or configuration is restored to its expected state and is accessible in the system interface
8. In an authorized isolated test environment, execute TEST-006: Given: an administrator requests backup status for multiple campus locations When: the administrator performs a cross-location status query Expected: the named items are returned in the requested format and are available for examination
9. In an authorized isolated test environment, execute TEST-007: Given: system detects a failed or incomplete backup operation When: the system initiates detection of a failed or incomplete backup operation Expected: the system reports the failure with identifiable metadata
10. In an authorized isolated test environment, execute TEST-008: Given: system identifies an alert condition for security or operational risk When: the system initiates the alert condition Expected: the requested alert exists in the designated channel
11. In an authorized isolated test environment, execute TEST-009: Given: system records backup and restore activities When: a backup or restore operation is completed Expected: The system records backup and restore activities when a backup or restore operation is completed, as evidenced by external test capture of the inspection record.
12. In an authorized isolated test environment, execute TEST-010: Given: user attempts to perform backup administration When: the user initiates an administrative action Expected: the system denies access if the user role lacks backup administration privileges
13. In an authorized isolated test environment, execute TEST-011: Given: system integrates with existing university storage and identity services using a controlled exchange When: the system initiates a connection request to the external service Expected: the external service responds with an authenticated session token and confirms successful path establishment
14. In an authorized isolated test environment, execute TEST-012: Given: A draft acceptance plan for this quality goal: the system shall be secure against unauthorized access and data leakage during backup and recovery operations. When: An analyst inspects or reviews that acceptance plan to determine whether measurable targets are defined or remain TBD/uncertain. Expected: The plan's measurable targets remain TBD or unconfirmed, as no specific metrics or thresholds for security assurance are provided in the current draft.
15. In an authorized isolated test environment, execute TEST-013: Given: A draft acceptance plan for this quality goal: the system shall remain secure during backup and recovery operations. When: An analyst inspects or reviews that acceptance plan to determine whether it includes measurable targets for security assurance. Expected: The plan's measurable targets remain TBD or unconfirmed; no specific metric, threshold, or test case is defined in the current draft.
16. In an authorized isolated test environment, execute TEST-014: Given: A draft acceptance plan for this quality goal: the system shall remain reliable during backup and recovery operations. When: An analyst inspects or reviews that acceptance plan to determine whether it includes measurable targets for reliability assurance. Expected: The plan's measurable targets remain TBD or unconfirmed; no specific metric, threshold, or test case is defined in the current draft.
17. In an authorized isolated test environment, execute TEST-015: Given: A draft acceptance plan for this quality goal: the system shall remain scalable during backup and recovery operations. When:

An analyst inspects or reviews that acceptance plan to determine whether it includes measurabletargets for scalability assurance. Expected: The plan's measurable targets remain TBD orunconfirmed; no specific metric, threshold, or test case is defined in the current draft.

18. In an authorized isolated test environment, execute TEST-016: Given: A draft acceptance plan for this quality goal: the system shall remain manageable during backup and recovery operations. When: An analyst inspects or reviews that acceptance plan to determine whether it includes measurable targets for manageability assurance. Expected: The plan's measurable targets remain TBD or unconfirmed; no specific metric, threshold, or test case is defined in the current draft.
19. Collect stakeholder feedback; agree its cadence with a human reviewer before scheduling cycles.
20. Return accepted stakeholder feedback and observed defects to the backlog and plan the next iteration.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Requirement-linked implementation increment
2. Recorded test evidence
3. Reviewed feedback and next-iteration backlog

Entry criteria

1. Prior work, source scope and the reviewed test plan are available for iteration planning.

Exit criteria

1. A human reviewer has assessed the increment, test evidence and feedback; unresolved changes return to the next iteration.

Security checks

1. Review implementation and acceptance evidence against FR-008: The system shall generate security and operational alerts.
2. Review implementation and acceptance evidence against NFR-001: The system shall be secure.
3. Review implementation and acceptance evidence against NFR-002: The system shall remain secure.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; FR-004; FR-005; FR-006; FR-007; FR-008; FR-009; FR-010; FR-011; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

Name: Release and handoff

Phase type: release

Activities

1. Review the exact requirements baseline, implementation links and verification evidence before a release decision.
2. Carry unresolved source decisions into handoff; generated text is not human approval.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Requirements and verification evidence package
2. Pending human release decision

3. Outstanding-decision record

Entry criteria

1. Verification evidence and unresolved decisions are available for human review.

Exit criteria

1. A human has recorded the release/handoff decision for the reviewed content.

Security checks

1. Review implementation and acceptance evidence against FR-008: The system shall generate security and operational alerts.
2. Review implementation and acceptance evidence against NFR-001: The system shall be secure.
3. Review implementation and acceptance evidence against NFR-002: The system shall remain secure.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; FR-004; FR-005; FR-006; FR-007; FR-008; FR-009; FR-010; FR-011; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

Unknown factors: budget; delivery; failure_consequences; feedback; legacy; regulation; schedule; stability; uncertainty; verification

SDLC rankings and decision evidence

Policy: sequential-sdlc-2.0

Generation profile: sequential-sdlc-generation-2.1

Factors

Name: security_assurance

Value: integrated security review

Evidence ids: E-002

Basis: derived

Rationale: Security decisions benefit from security review and tests throughout delivery.

Name: operational_validation

Value: monitoring feedback

Evidence ids: E-002; E-003

Basis: derived

Rationale: Monitoring and alert behavior needs validation with operational feedback; this does not establish continuous deployment or a release frequency.

Name: deployment_scope

Value: staged rollout opportunity

Evidence ids: E-002

Basis: derived

Rationale: Multiple deployment locations offer a staged validation and rollout opportunity; the rollout plan remains proposed.

Name: integration_assurance

Value: integration validation

Evidence ids: E-004

Basis: derived

Rationale: Connected systems need integration verification before broader rollout; their age, technology and technical uncertainty remain unspecified.

Ranked candidates

Model: Incremental

Score: 66

Reasons: Derived operational_validation (+6 points): Monitoring and alert behavior needs validation with operational feedback; this does not establish continuous deployment or a release frequency. (E-002, E-003); Derived deployment_scope (+12 points): Multiple deployment locations offer a staged validation and rollout opportunity; the rollout plan remains proposed. (E-002); Derived integration_assurance (+8 points): Connected systems need integration verification before broader rollout; their age, technology and technical uncertainty remain unspecified. (E-004)

Model: Hybrid

Score: 65

Reasons: Derived security_assurance (+8 points): Security decisions benefit from security review and tests throughout delivery. (E-002); Derived operational_validation (+5 points): Monitoring and alert behavior needs validation with operational feedback; this does not establish continuous deployment or a release frequency. (E-002, E-003); Derived deployment_scope (+8 points): Multiple deployment locations offer a staged validation and rollout opportunity; the rollout plan remains proposed. (E-002); Derived integration_assurance (+4 points): Connected systems need integration verification before broader rollout; their age, technology and technical uncertainty remain unspecified. (E-004)

Model: DevSecOps

Score: 63

Reasons: Derived security_assurance (+12 points): Security decisions benefit from security review and tests throughout delivery. (E-002); Derived operational_validation (+8 points): Monitoring and alert behavior needs validation with operational feedback; this does not establish continuous deployment or a release frequency. (E-002, E-003); Derived deployment_scope (+3 points): Multiple deployment locations offer a staged validation and rollout opportunity; the rollout plan remains proposed. (E-002)

Model: Spiral

Score: 51

Reasons: Derived security_assurance (+5 points): Security decisions benefit from security review and tests throughout delivery. (E-002); Derived integration_assurance (+6 points): Connected systems need integration verification before broader rollout; their age, technology and technical uncertainty remain unspecified. (E-004)

Model: Agile

Score: 40

Reasons: No differentiating input evidence

Model: V-Model

Score: 40

Reasons: No differentiating input evidence

Model: Waterfall

Score: 40

Reasons: No differentiating input evidence

Evidence register

Id: E-001

Source: description

Quote: Design a secure backup and recovery management system for a university that protects critical data from academic departments, administrative systems, research platforms, and shared services.

Id: E-002

Source: description

Quote: The system should allow authorized administrators to define backup policies, initiate and monitor backup jobs, restore approved data, and review backup status across multiple campus locations.

Id: E-003

Source: description

Quote: It should detect failed or incomplete backup operations, generate security and operational alerts, maintain audit logs of backup and restore activities, and restrict backup administration based on user roles.

Id: E-004

Source: description

Quote: The platform should integrate with existing university storage and identity services and remain secure, reliable, scalable, and manageable.

Source references

kb-c1e0239af9dd46bd:1 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 1

Page or section: text chunk 2

Relevance score: 0.7073

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-92ffa375cd05a8f5:6 - 03_OWASP_CheatSheets/081_OWASP_Multifactor_Authentication_Cheat_Sheet.md

Source document id: kb-92ffa375cd05a8f5

Chunk index: 6

Page or section: text chunk 7

Relevance score: 0.7102

Sha256: 0c1de91561bd8b449f5a27033e799c6928df3f24cb48a965994f615671612490

kb-ecd06e0e85ebe229:15 - 03_OWASP_CheatSheets/038_OWASP_Authorization_Cheat_Sheet.md

Source document id: kb-ecd06e0e85ebe229

Chunk index: 15

Page or section: text chunk 16

Relevance score: 0.6887

Sha256: e78a10eb9759a1fc7593ba1fc89a05b017e7e795ac16ad5496bdba86138d891f

kb-c1e0239af9dd46bd:2 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 2

Page or section: text chunk 3

Relevance score: 0.6745

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-c1e0239af9dd46bd:11 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 11

Page or section: text chunk 12

Relevance score: 0.6886

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-ecd06e0e85ebe229:16 - 03_OWASP_CheatSheets/038_OWASP_Authorization_Cheat_Sheet.md

Source document id: kb-ecd06e0e85ebe229

Chunk index: 16

Page or section: text chunk 17

Relevance score: 0.6765

Sha256: e78a10eb9759a1fc7593ba1fc89a05b017e7e795ac16ad5496bdba86138d891f

Validation findings

Code: human_semantic_review_required

Severity: warning

Section: document

Message: Review completeness, actor permissions, contradictions and citation applicability; schema and link checks do not prove semantic correctness.

Human review history

None recorded.

AI drafting record

Requested model: cybersrs-qwen3-4b-ft:latest

Actual models used: cybersrs-qwen3-4b-ft:latest; qwen3:4b-instruct-2507-q4_K_M

Section models

Introduction: cybersrs-qwen3-4b-ft:latest

Functional requirements: cybersrs-qwen3-4b-ft:latest; qwen3:4b-instruct-2507-q4_K_M

Non-functional requirements: cybersrs-qwen3-4b-ft:latest

Compliance and policy applicability matrix: cybersrs-qwen3-4b-ft:latest

Architecture: cybersrs-qwen3-4b-ft:latest

Threats: cybersrs-qwen3-4b-ft:latest

Risks: cybersrs-qwen3-4b-ft:latest

User stories and use cases: cybersrs-qwen3-4b-ft:latest

Verification and acceptance: cybersrs-qwen3-4b-ft:latest

Glossary: cybersrs-qwen3-4b-ft:latest

SDLC recommendation and workflow: qwen3:4b-instruct-2507-q4_K_M

Prompt version: sequential-1.36

Requirements traceability matrix

Requirement	Evidence / sources	Stories / tests / SDLC
FR-001	E-001	TEST-001, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-002	E-002	UC-001, TEST-002, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-003	E-002	UC-002, TEST-003, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-004	E-002	UC-003, TEST-004, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-005	E-002	UC-004, TEST-005, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-006	E-002	UC-005, TEST-006, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-007	E-003	TEST-007, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-008	E-003	TEST-008, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-009	E-003	TEST-009, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-010	E-003	TEST-010, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-011	E-004	TEST-011, Requirements and verification planning, Incremental delivery cycle, Release and handoff
NFR-001	E-001	TEST-012, Requirements and verification planning, Incremental delivery cycle, Release and handoff
NFR-002	E-004	TEST-013, Requirements and verification planning, Incremental delivery cycle, Release and handoff
NFR-003	E-004	TEST-014, Requirements and verification planning, Incremental delivery cycle, Release and handoff

Requirement	Evidence / sources	Stories / tests / SDLC
NFR-004	E-004	TEST-015, Requirements and verification planning, Incremental delivery cycle, Release and handoff
NFR-005	E-004	TEST-016, Requirements and verification planning, Incremental delivery cycle, Release and handoff

Model: cybersrs-qwen3-4b-ft:latest + qwen3:4b-instruct-2507-q4_K_M; prompt: sequential-1.36